

Complete Beginner's Guide: Connecting Google Services to n8n

Setting up Google services with n8n can seem overwhelming at first, but it's actually straightforward once you understand the key differences between n8n Cloud and self-hosted setups. This guide will walk you through everything you need to know, from the simplest one-click connections to more advanced configurations.

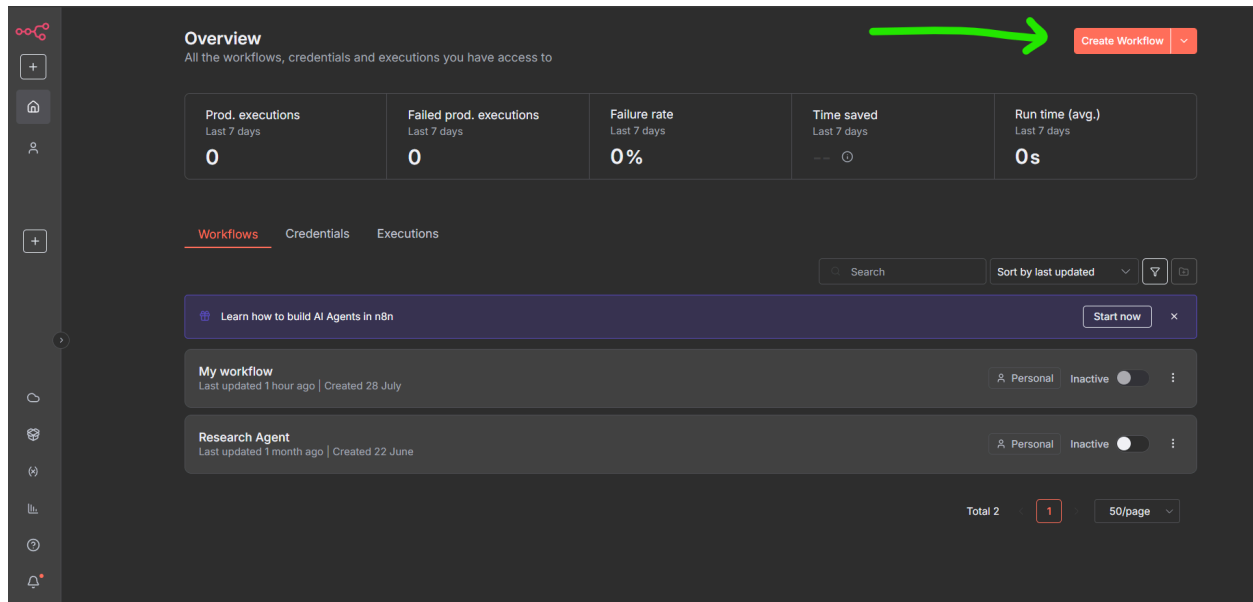
Quick Overview: What Works Where

Service	n8n Cloud	Self-Hosted n8n
Gmail	✅ One-click "Sign in with Google"	❌ Requires Google Cloud setup
Google Calendar	✅ One-click "Sign in with Google"	❌ Requires Google Cloud setup
Google Sheets	✅ One-click "Sign in with Google"	❌ Requires Google Cloud setup
Google Drive	❌ Requires Google Cloud setup	❌ Requires Google Cloud setup
Google Docs	❌ Requires Google Cloud setup	❌ Requires Google Cloud setup

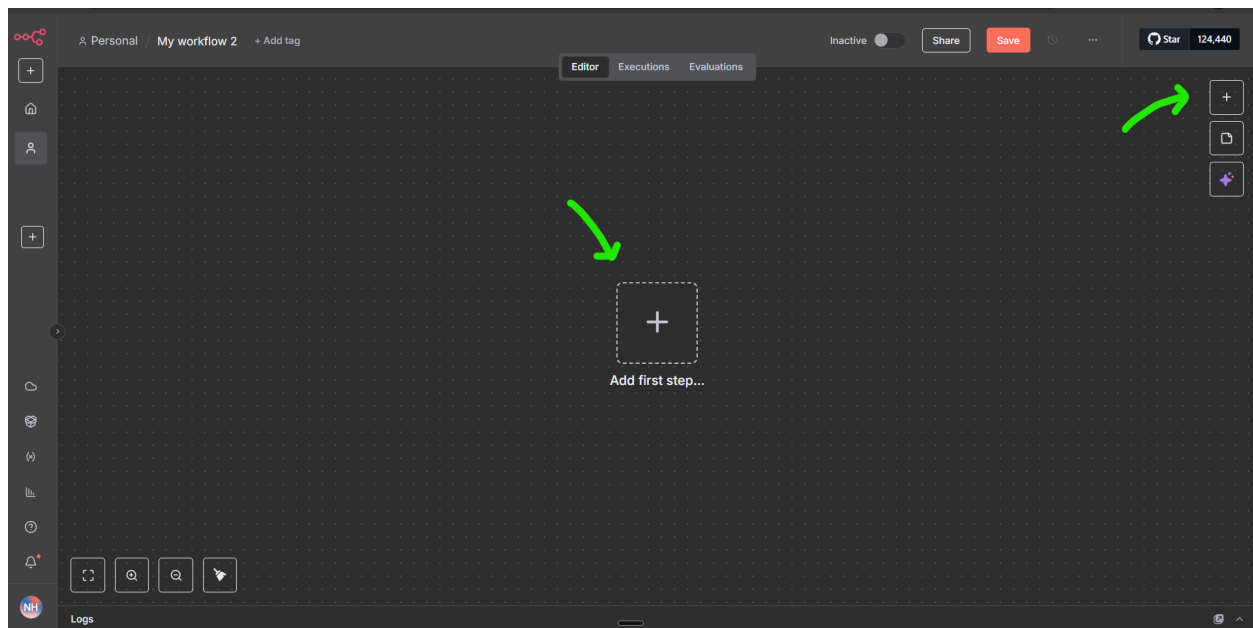
Part 1: Using n8n Cloud (The Easy Way)

If you haven't set up n8n yet, set up an account on their cloud. It is the simplest setup and you can start with a 14 day free trial: <https://n8n.partnerlinks.io/22crlu8afq5r>

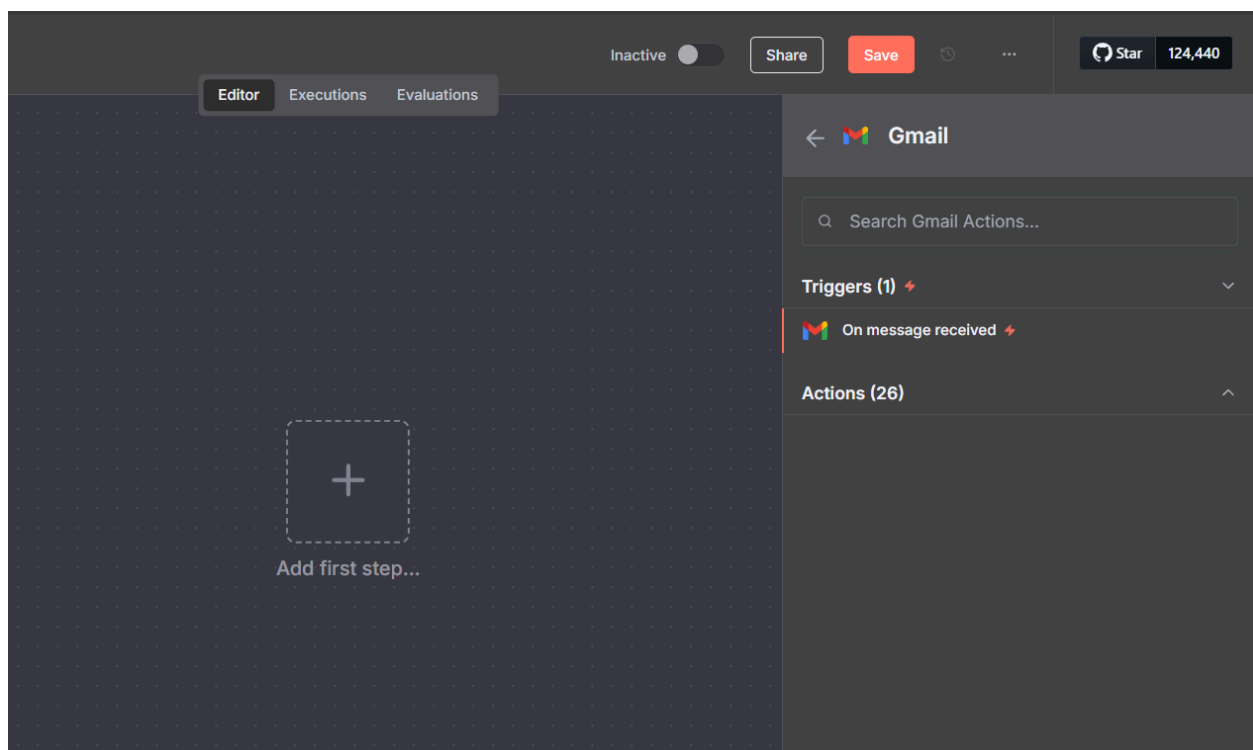
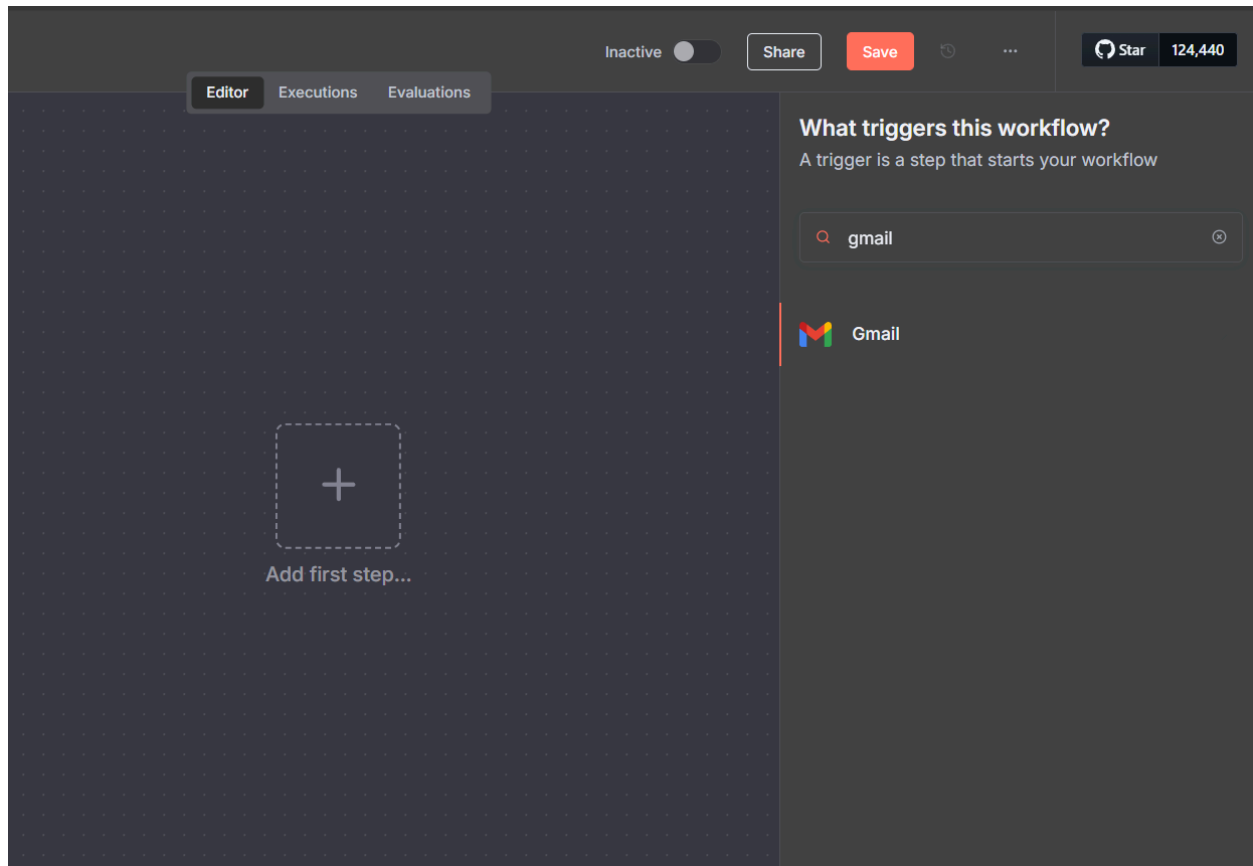
In your new n8n instance, create a new workflow:



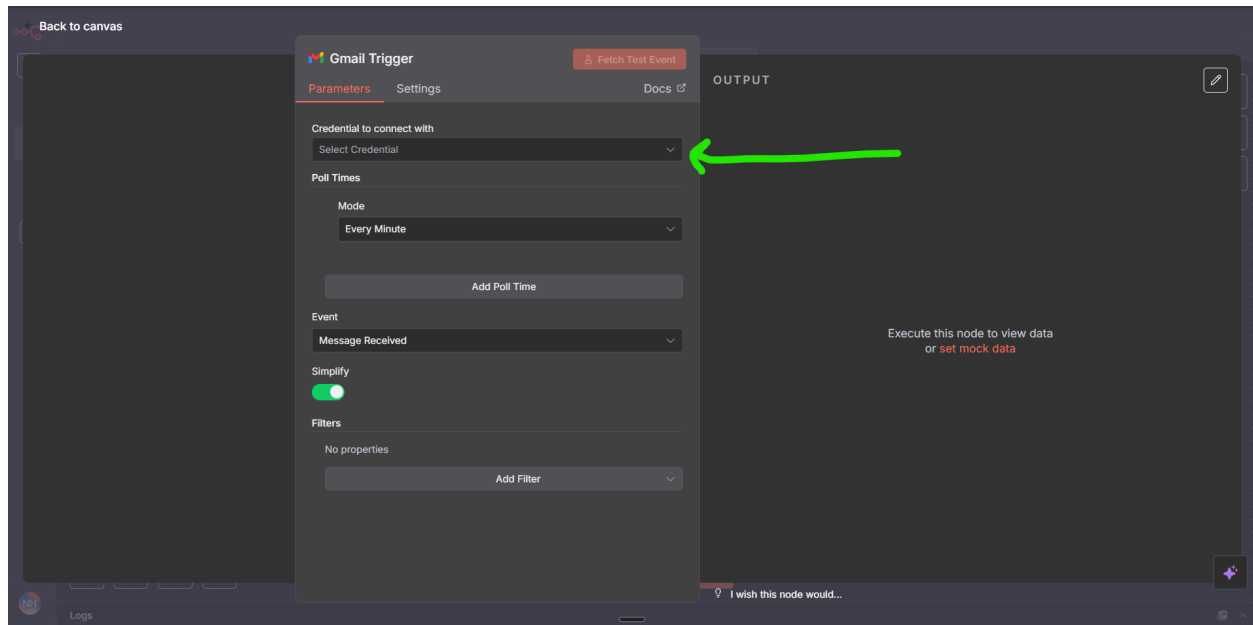
Click on “Add first step” or the plus icon near the top right to add a google node to connect to. Let’s start with Gmail.



Type “Gmail” in the search bar and grab a Trigger, “On message received. (You can tell it is a trigger rather than an action because of the orange lightning bolt next to it.



Whenever you want to connect to an app, you have to create a credential, which can be found near the top of the configuration panel.



Gmail, Calendar, Sheets - One-Click Setup

If you're using n8n Cloud, connecting Gmail and Google Calendar is incredibly simple:

1. Add a Google node to your workflow
2. Click "Create new credential"
3. Click "Sign in with Google"
4. Choose your Google account and grant permissions
5. Done! Your credential is ready to use

Why this works: n8n Cloud comes with pre-configured Google OAuth credentials that already have Gmail and Calendar APIs enabled. You're using n8n's verified Google Cloud project, so no setup is required on your end.

You may have seen other videos on YouTube for Google setups where you have to create a Google Cloud project and get a client ID and a secret for every Google app. n8n Cloud has made it easier for these nodes, but with Google Drive and Google Docs, it does require a cloud setup, which you can do for free.


Gmail account
 Gmail OAuth2 API

Connection

Sharing

Details

Connect using *

☒ OAuth2 (recommended)

☐ Service Account


 Sign in with Google

ⓘ Enterprise plan users can pull in credentials from external vaults. [More info](#)


Google Calendar account
 Google Calendar OAuth2 API

Connection

Sharing

Details


 Sign in with Google

ⓘ Enterprise plan users can pull in credentials from



Google Sheets account

Google Sheets OAuth2 API

Connection

Sharing

Details

Connect using *

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Google Drive account

Google Drive OAuth2 API

Connection

Sharing

Details

Need help filling out these fields? [Open docs](#)

Connect using *

☒ OAuth2 (recommended)
 ☐ Service Account

[Ask Assistant](#) for setup instructions

OAuth Redirect URL

`https://oauth.n8n.cloud/oauth2/callback`

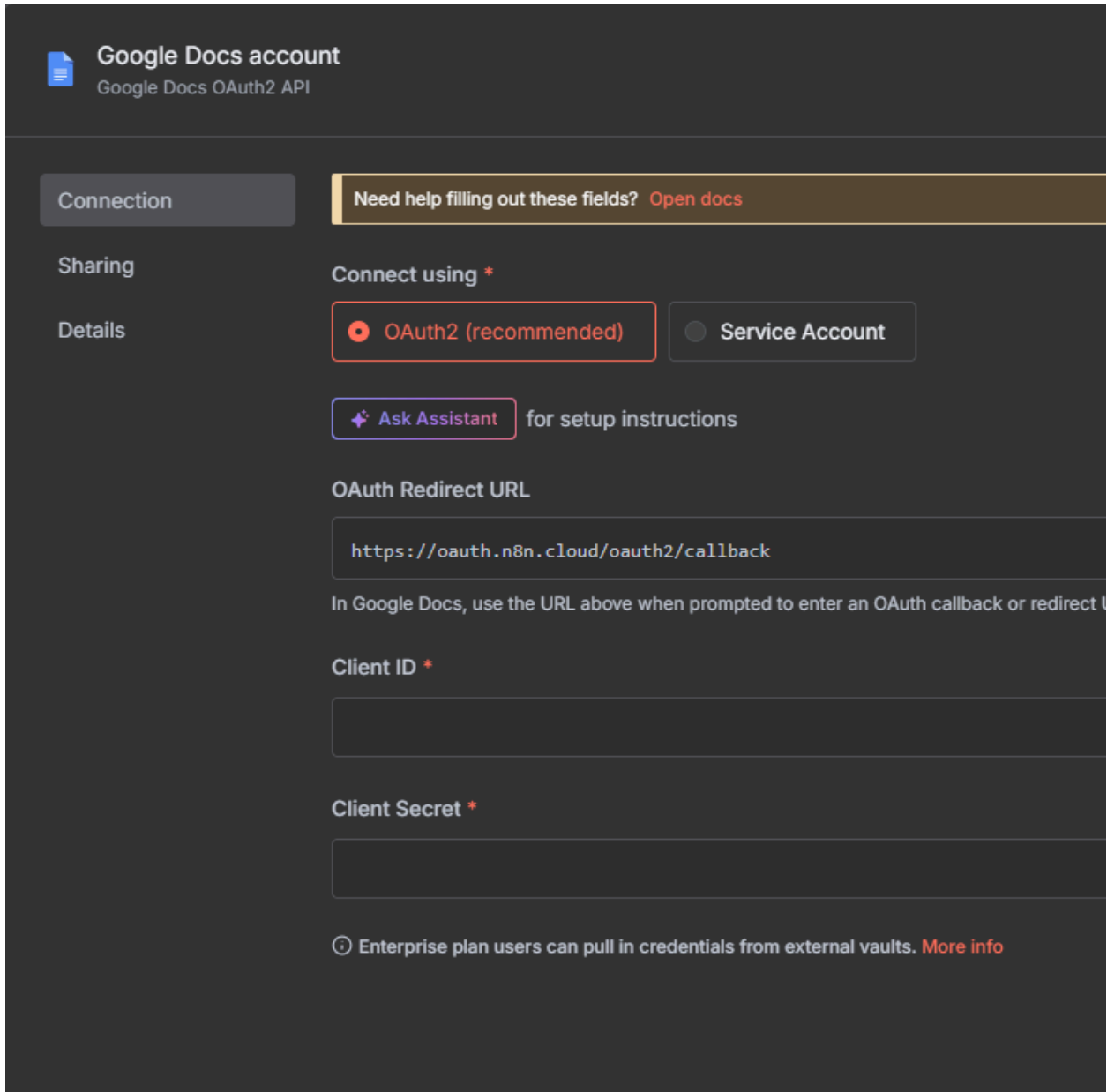
In Google Drive, use the URL above when prompted to enter an OAuth callback or redirect URL

Client ID *

Client Secret *

Make sure that you have enabled the Google Drive API in the Google Cloud Console. [More info.](#)

ⓘ Enterprise plan users can pull in credentials from external vaults. [More info](#)



The screenshot shows the 'Google Docs account' setup page for the 'Google Docs OAuth2 API'. It features a sidebar with 'Connection', 'Sharing', and 'Details' tabs. The 'Connection' tab is active, displaying a 'Connect using' section with 'OAuth2 (recommended)' selected and 'Service Account' as an alternative. Below this is a link to 'Ask Assistant' for setup instructions. The 'OAuth Redirect URL' field is populated with 'https://oauth.n8n.cloud/oauth2/callback'. A note instructs users to use this URL in Google Docs. The 'Client ID' and 'Client Secret' fields are empty. A footer note mentions that enterprise plan users can pull in credentials from external vaults.

Google Docs account
Google Docs OAuth2 API

Connection

Need help filling out these fields? [Open docs](#)

Sharing

Details

Connect using *

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[Ask Assistant](#) for setup instructions

OAuth Redirect URL

`https://oauth.n8n.cloud/oauth2/callback`

In Google Docs, use the URL above when prompted to enter an OAuth callback or redirect URL

Client ID *

Client Secret *

Enterprise plan users can pull in credentials from external vaults. [More info](#)

Part 2: Setting Up Your Google Cloud Project

Whether you're on n8n Cloud (for Drive/Docs) or self-hosting, you'll need to create a Google Cloud project. Here's the step-by-step process:

Step 1: Create a Google Cloud Project

1. Go to console.cloud.google.com
2. Click "Select a project" → "New Project"

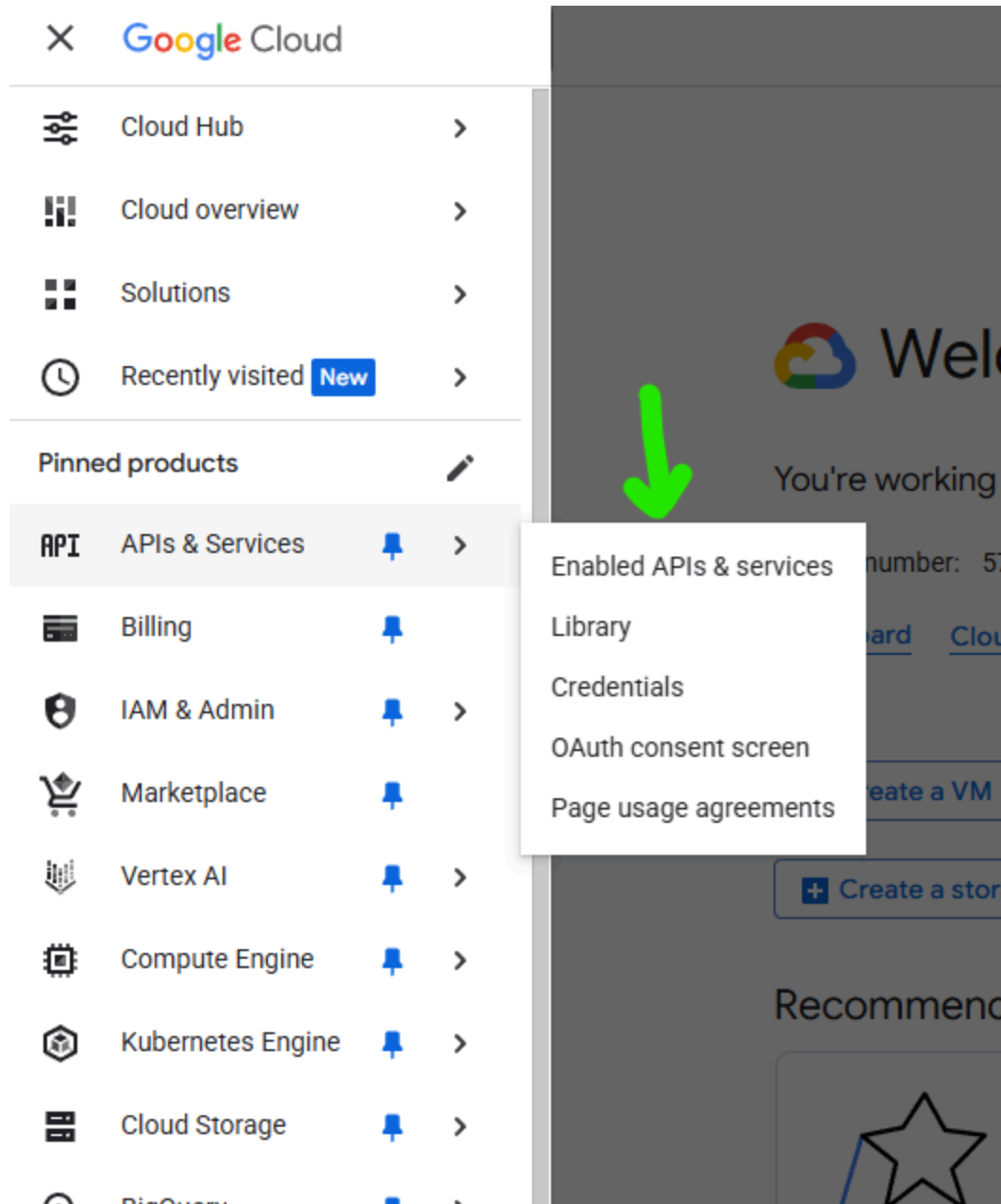
3. Name your project (e.g., "n8n-automations")
4. Click "Create"

Step 2: Enable Required APIs

Navigate to APIs & Services → Library and enable these APIs:

- Gmail API (if self-hosting)
- Google Calendar API (if self-hosting)
- Google Drive API (required for all file operations)
- Google Sheets API (if self-hosting)
- Google Docs API (if using Docs)
- Cloud Resource Manager API (needed for Drive file listing)

You can use the search bar to find these APIs, then it is as simple as clicking enable



Google Cloud Demo

Search (/) for resources, docs, products, and more

APIs & Services

Enabled APIs & services

Library

Credentials

OAuth consent screen

Page usage agreements

Traffic

No data is available for the selected time frame.

Errors

No data is available for the selected time frame.

UTC-5 6:00 PM Jul 28 6:00 AM

Filter Filter

Product details



Gmail API

[Google Enterprise API](#)

View and manage Gmail mailbox data.

Manage

[Try this API](#)



API Enabled

Step 3: Configure OAuth Consent Screen

1. Go to APIs & Services → OAuth consent screen
2. Choose User Type:
 - Internal: Only for Google Workspace users (tokens never expire)
 - External: For any Google account users (token expired every 7 days in test mode)
3. Fill in required fields:
 - App name (e.g., "n8n Automations")
 - Support email
 - Developer contact information
4. Important: If you choose External, click "Publish App" to avoid the 7-day token expiration issue.
 - OAuth consent screen → Audience → Publish app

✓ App Information

2 Audience

☐ Internal ?

Only available to users within your organization. You will not need to submit your app for verification. [Learn more about user type](#)

☐ External ?

Available to any test user with a Google Account. Your app will start in testing mode and will only be available to users you add to the list of test users. Once your app is ready to push to production, you may need to verify your app. [Learn more about user type](#)

Next



Google Cloud



Demo



Google Auth Platform / Audience



Overview



Branding



Audience



Clients



Data Access



Verification Center

Audience

Testing

Publish app

User type

External ?

Make internal

5. If you're leaving your app in test mode, make sure to add the email you want to use as a test user in the audience section of your OAuth consent screen. **Only emails that you add have access to this project until your app is in production mode.**

Test users

[+ Add users](#)



Filter Enter property name or value



User information

[redacted]@gmail.com



Step 4: Create OAuth Credentials

1. Go to APIs & Services → Credentials
2. Click "Create Credentials" → "OAuth 2.0 Client ID"
3. Select "Web application"
4. Add your Authorized redirect URI:
 - n8n Cloud: Copy the redirect URL from your n8n credential setup

Google Drive account
Google Drive OAuth2 API

Need help filling out these fields? [Open docs](#)

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Sharing

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OAuth Redirect URL

`https://oauth.n8n.cloud/oauth2/callback`

In Google Drive, use the URL above when prompted to enter an OAuth callback or redirect URL

Client ID *

Client Secret *

Make sure that you have enabled the Google Drive API in the Google Cloud Console. [More info](#)

☐ Enterprise plan users can pull in credentials from external vaults. [More info](#)

Save

-
- Self-hosted: Use your domain + `/rest/oauth2-credential/callback`
- If Local Hosted: Configure a public URL via a tunnel

[▶ How to Set Up a Cloudflare Tunnel for Local n8n \(2025\)](#)

5. Save your Client ID and Client Secret

Step 5: Add Credentials to n8n

1. In n8n, go to any Google node → "Create new credential"
2. Select "OAuth2"
3. Paste your Client ID and Client Secret
4. Click "Sign in with Google"
5. Grant the requested permissions

Part 3: Self-Hosted n8n Setup

Basic Requirements

When self-hosting n8n, you need:

- A publicly accessible domain with HTTPS
- Proper environment variables configured
- A tunnel service (if running locally)

Setting Up Local Development with Tunnels

If you're running n8n locally for development, you'll need to expose it to the internet for Google OAuth callbacks:

In this video, I show you how to set up a Cloudflare Tunnel for Local n8n

 [How to Set Up a Cloudflare Tunnel for Local n8n \(2025\)](#)

Configure n8n Environment Variables

Set these environment variables when starting n8n:

bash

For Docker

```
docker run -it --rm --name n8n -p 5678:5678 \
  -e WEBHOOK_URL=https://abc123.ngrok.io \
  -e N8N_HOST=abc123.ngrok.io \
  -e N8N_PROTOCOL=https \
  n8nio/n8n
```

For npm

```
export WEBHOOK_URL=https://abc123.ngrok.io
export N8N_HOST=abc123.ngrok.io
export N8N_PROTOCOL=https
npx n8n
```

Important: Update these URLs each time you restart ngrok, as the free version generates new URLs.

Production Self-Hosted Setup

For production deployments:

1. Use a real domain: Register a domain and set up DNS
2. Configure SSL: Set up HTTPS with Let's Encrypt or similar

3. Set environment variables:
4. `bash`

```
N8N_HOST=n8n.yourdomain.com
N8N_PROTOCOL=https
WEBHOOK_URL=https://n8n.yourdomain.com
N8N_EDITOR_BASE_URL=https://n8n.yourdomain.com
```

- 5.

Part 4: Common Issues and Solutions

Authentication Problems

Issue: "Token has been expired or revoked" every 7 days

Solution: Ensure your Google Cloud app is published to "Production" status, not left in "Testing".

Issue: "Unable to sign without access token"

Solution:

- Upgrade to n8n v1.75+
- Set `N8N_PROXY_HOPS=1` if using a proxy
- Ensure HTTPS is properly configured

Issue: "redirect_uri_mismatch"

Solution: Verify that the redirect URI in Google Cloud Console exactly matches what n8n shows in the credential setup.

Connection Issues

Issue: Google OAuth shows "This app isn't verified"

Solution: This is normal for unpublished external apps. Click "Advanced" → "Go to [your app]" to proceed.

Issue: "Localhost refused to connect"

Solution: Google can't reach localhost from the internet. Use a tunnel service or deploy to a public domain.

Issue: Credentials work briefly then fail

Solution: Check if your app is still in "Testing" mode. Publish it to "Production" or set user type to "Internal".

Self-Hosted Specific Issues

Issue: Webhooks not working with tunnels

Solution: Ensure your tunnel URL is set in all relevant environment variables (`WEBHOOK_URL`, `N8N_HOST`, etc.).

Issue: SSL certificate problems

Solution: Include `https://` in your environment variables and ensure your SSL certificates are valid.

Frequently Asked Questions

Q: Why can I connect Gmail easily but not Google Drive on n8n Cloud?

A: n8n Cloud includes pre-configured credentials for Gmail and Calendar, but Drive/Docs require broader permissions that need your own Google Cloud project.

Q: Do I need to pay for Google Cloud?

A: No, the Google Cloud free tier is sufficient for most n8n integrations. You only pay if you exceed the generous free quotas.

Q: Can I use the same credentials for multiple Google services?

A: Yes! Once you create OAuth credentials with the right scopes, you can reuse them across Gmail, Drive, Sheets, Docs, and Calendar nodes.

Q: My tokens keep expiring - what's wrong?

A: This usually means your Google Cloud app is in "Testing" mode. Change it to "Production" or set the user type to "Internal" if you're using Google Workspace.

Q: Can I use n8n locally without a domain?

A: For development, yes - use ngrok or similar tunnel services. For production, you'll need a real domain with SSL.

Q: What if I'm behind a corporate firewall?

A: Ensure your firewall allows connections to `oauth2.googleapis.com` and `accounts.google.com`. You may need to whitelist these domains.

Getting Started Checklist

For n8n Cloud users:

- Try Gmail/Calendar with one-click signin first
- Create Google Cloud project for Drive/Docs/Sheets
- Enable required APIs
- Set up OAuth consent screen and publish app
- Create OAuth credentials and add to n8n

For self-hosted users:

- Set up domain with SSL OR configure tunnel for development
- Configure environment variables
- Create Google Cloud project
- Enable all required APIs
- Set up OAuth consent screen and publish app
- Create OAuth credentials and add to n8n
- Test webhook functionality

Remember: Start simple with one service, get it working, then expand to others. The setup process is the same for each Google service - you just need to enable the appropriate APIs and include the right scopes in your OAuth setup.